



PROJECT

NATIONAL BOARD-CERTIFIED TEACHER (NBCT) TRENDS ANALYSIS

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ABBREVIATIONS

ALSDE	:	Alabama State Department of Education
C100	:	Huntsville Committee of 100
CEO	:	Chief Executive Officer
NAEP	:	National Assessment of Educational Progress
NBCT	:	National Board-Certified Teachers
NBPTS	:	National Board for Professional Teaching Standards
NCES	:	National Center for Education Statistics
RYH	:	Raise Your Hand

EXECUTIVE SUMMARY

This report analyzed the distribution of National Board-Certified Teachers (NBCT) in Alabama and its correlation with socio-economic factors and student performance. The report provides insights into the relationship between the distribution of NBCT's, student performance in Alabama and the factors that affect the distribution of NBCT's across different regions. These insights contribute towards Huntsville Committee 100's "Raise Your Hand" campaign.

The analysis found a positive correlation between the number of NBCTs in a state and the State average NAEP scores. The correlation was stronger in lower grade performance compared to higher grade scores.

Alabama counties have an uneven distribution of NBCTs. The total number of NBCT's in Alabama is 2,978, which corresponds to a total NBCT to Teacher Ratio of 2.98%. The maximum NBCT's to Teacher Ratio is 19.9%, found in Jefferson County, and the minimum NBCT's to Teacher Ratio is 0, found in Crenshaw and Henry counties. Madison County has an NBCT's to Teacher Ratio of 8.5%, which is above the state average of 5.2%.

When it comes to student performance, the average percentage of Grade 3 students below grade level for all counties in Alabama is about 24.9%. The county with the highest percentage of Grade 3 students below grade level is Wilcox with 57.66%. The county with the lowest percentage of Grade 3 students below grade level is Cleburne with 5.88%. Madison County has 13.96% of grade 3 students below grade level.

The proportion of students receiving free and reduced lunch per county tends to correlate strongly with proportion of students below the grade 3 level. This shows correlation between student performance and socioeconomic factors. Comparing this with the counties proportion of NBCT shows that counties with higher proportion tend to have better performance, despite the counties

socio economic factors. Counties such as Jefferson, that have 49% of students on free and reduced lunch and the highest proportion of NBCTs report better performance.

These results indicate that an investment by Huntsville/Madison County in increasing the proportion of NBCT's should therefore be strongly advocated for and supported, to improve performance in schools.

INTRODUCTION

Background Statement

The Huntsville Committee of 100 (C100) launched the Raise Your Hand (RYH) campaign in February 2021 aimed at raising funds to support teachers from Huntsville City Schools to pursue NBCT certification. Funds from the campaign go towards adding 500 new National Board-Certified Teachers (NBCTs) across three school systems in Huntsville City, Madison City, and Madison County School systems.

To support this initiative, our team delved into collecting and analyzing data to gain a better understanding of students' performance in relation to NBCT's concentrations in Alabama. National Board Certification is an advanced teaching credential that is awarded by the National Board for Professional Teaching Standards (NBPTS). The certification process requires teachers to demonstrate their knowledge and skills in their subject area and teaching practices through rigorous assessments. Teachers who earn NBCT certification are recognized as accomplished educators who have met high standards for teaching effectiveness.

Alabama has a notable number of NBCT teachers, however, the distribution of NBCT's varies significantly across different regions of Alabama, and there is a need to understand the factors that affect the distribution of NBCT's and their impact on student performance.

This report aims to provide insights into the distribution of NBCT's in Alabama and their correlation with socio-economic factors and student performance. The report analyzes the available data on the distribution of NBCT's and socio-economic factors in Alabama to identify relationships and trends that can help policymakers and education stakeholders in developing strategies to improve academic performance among students.

Problem Statement

Our team worked with the Huntsville Committee of 100 (C100), led by the CEO, John Allen, to perform an analysis of NBCT's concentration within the public-school systems in Alabama and their correlation with socio-economic factors and student performance. The project was also aimed at providing insights into the relationship between the distribution of NBCT's and student performance in Alabama, as well as the factors that affect the distribution of NBCT's across different regions.

Project Objectives

The project objectives were as follows.

- To find the relationship between student performances in relation to the number of NBCT's.
- To compare student performances in relation to number of NBCT's certifications in Alabama.
- To explore the relationship among student performance, number of NBCT's in Alabama and other socio-economic factors.

The following datasets were used to address the above objectives:

- NBCT Teachers Data
- National NAEP Score Data
- Alabama NCES Public Schools Data
- Alabama Grade Three Reading Scores
- Poverty Status by School Enrollment Data
- Geographic Mobility Data

METHODOLOGY

The report analyzed data from the National Board for Professional Teaching Standards (NBPTS) and the Alabama State Department of Education (ALSDE) on the distribution of NBCT's and student performance in Alabama. The data was analyzed using data analytics tools and techniques to identify trends and correlations.

The team acquired data summaries for counties in Alabama from various data sources including previous data from the Committee of 100. The data was combined into fewer datasets, then ratios and proportions were calculated and summarized to counties in Alabama and states. Exploratory data analysis was used to gain insights and inform findings.

FINDINGS

National NBCT Landscape

The distribution of NBCT's across states in the country is uneven. NBCT's (2022) figures show that out of 130,493 certified teachers in the country, states in the east and southern part have a higher proportion of all the certified teachers in the country while states in the northern parts seem to have the lowest proportion of certified teachers. Jurisdictions with the lowest proportions of certified teachers include New Hampshire (0.029%), North Dakota (0.06%) and the District of Columbia (0.07%). Jurisdictions with the highest proportions of certified teachers include North Carolina (17.95%), Florida (10.4%), and Washington (9.03%). Alabama has 2.5% of all the certified teachers in the State and ranks at position 12. Below is a visual representation of the distribution of teachers across states in the US.

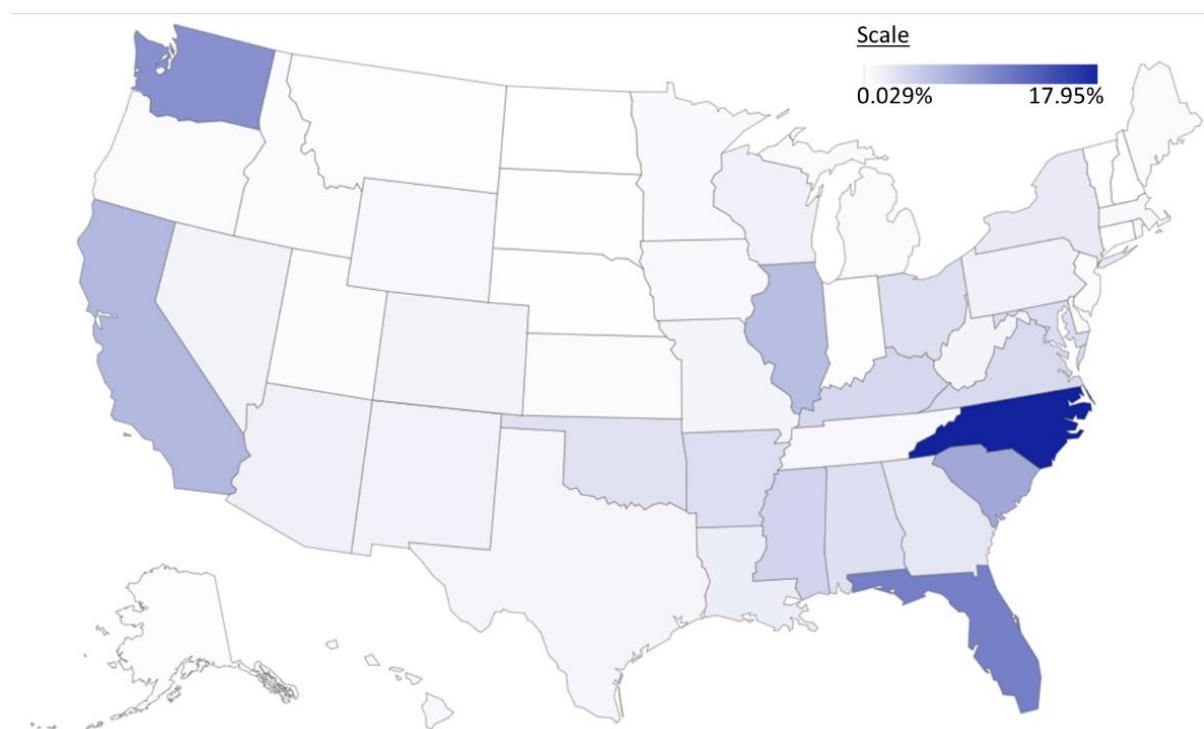


Figure: 1.0

Relationship Between Student Performance and Number of NBCT's

Findings show a positive correlation between the number of NBCT's and average NAEP scores for all subjects except Grade 8 Math in 2017. The findings show a stronger correlation between Grade 4 NAEP scores compared to Grade 8 NAEP scores as shown in Figure 2.0 below.

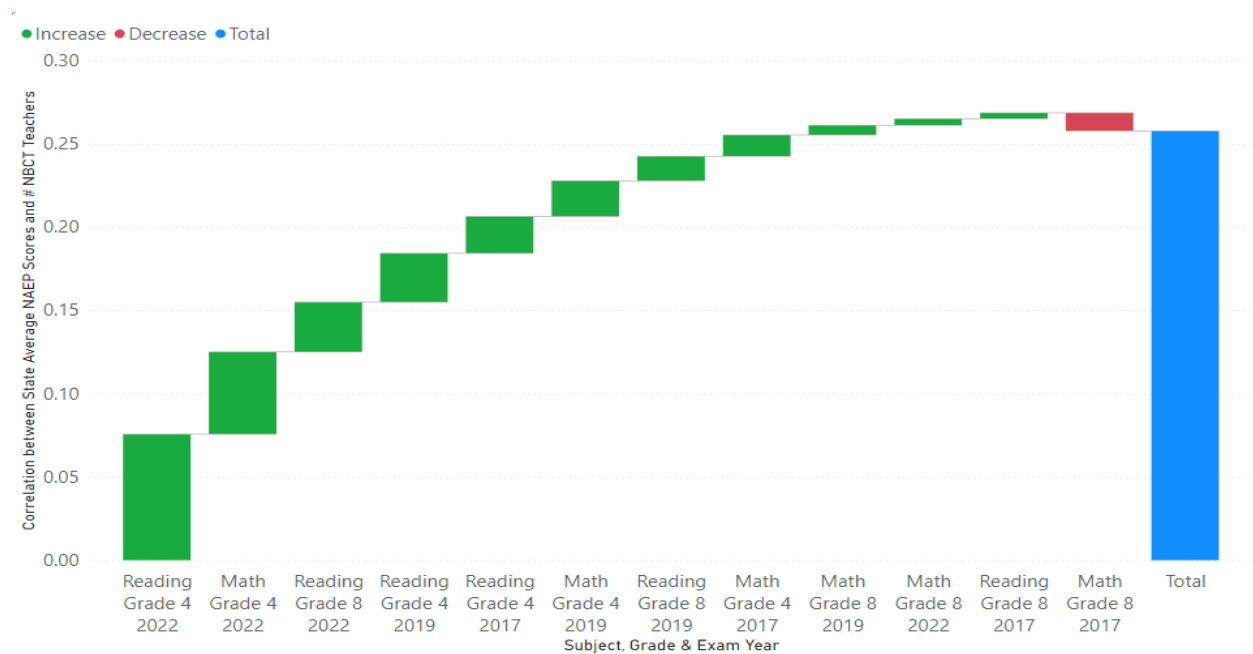


Figure 2.0

These findings indicate that state jurisdictions with more NBCT's have higher NAEP scores than states with fewer NBCT's. Increasing the number of NBCT's in schools would therefore lead to better performance among students, especially among students in lower grades. Findings indicate there is a positive correlation between grade 8 Math in all the years (2017, 2019 and 2022) and grade 8 reading in 2017 and 2019. This shows that states with a higher number of NBCT's have a bigger proportion of grade 8 students with below Basic NAEP scores in Math.

On the other hand, there is a negative correlation between the number of certified NBCT's and Grade 4 scores as well as Grade 8 reading in 2022. This shows that states with more NBCT's have smaller proportions of students with below Basic NAEP scores. This confirms the findings that having more NBCT's could have a bigger impact on student performance in earlier years of study than in later years.



Figure 3.0

Alabama Education Summary

The findings illustrate that rural schools have the highest proportion of both students and teachers, as well as the highest percentage of students on free and reduced lunch program. City schools

have the second-highest proportion of students and teachers, while suburb and town schools have the lowest proportions as illustrated in figure 4.0 below.

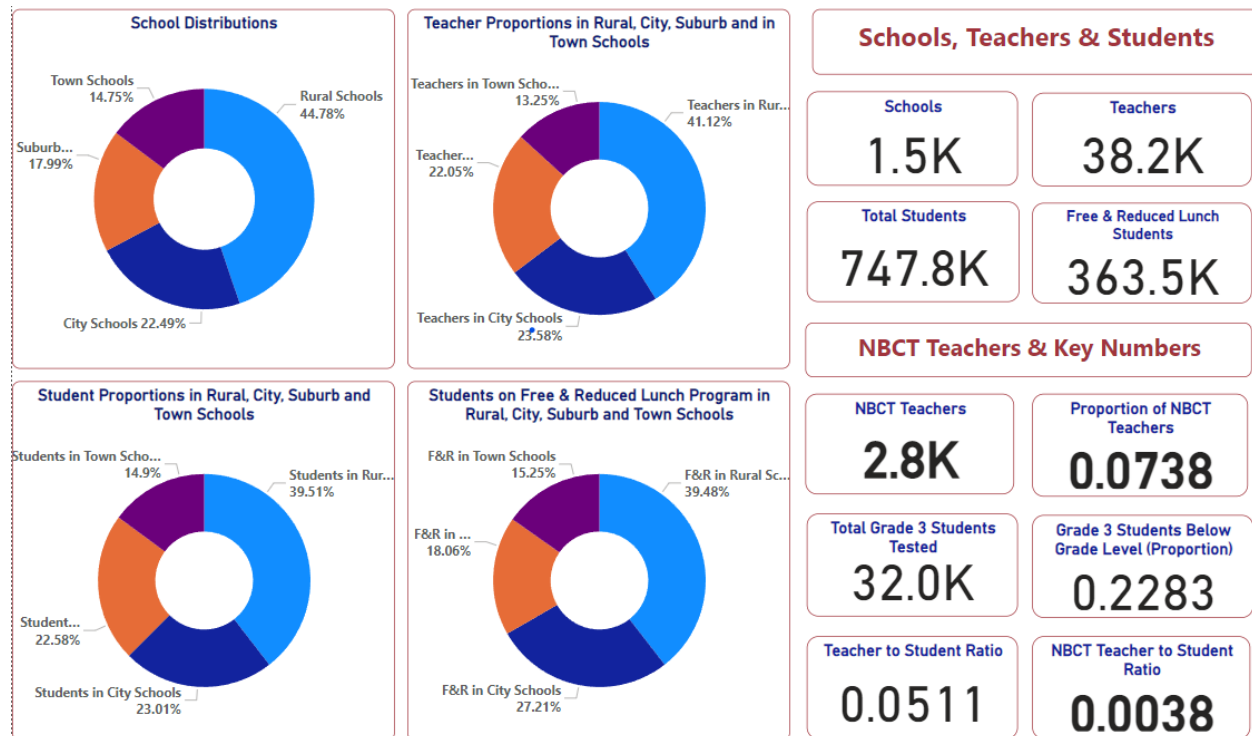


Figure 4.0

Madison County Summary

The chart below shows the percentage of schools in Madison that are located in three different areas: the city, suburbs, and rural areas indicating that the majority of schools in Madison, or 53.41%, are located in the city, followed by 29.55% in the suburbs, and 14.77% in rural areas. This suggests that Madison is primarily an urban area, with the highest concentration of schools located in the city.

The chart provides insights into the distribution of teachers across different types of schools in Madison where 45.51% of the teachers work in city schools, 35.98% in suburb schools, 14.8%

in rural schools, and 3.71% in town schools. The percentage of teachers in rural schools is relatively low compared to the other areas.

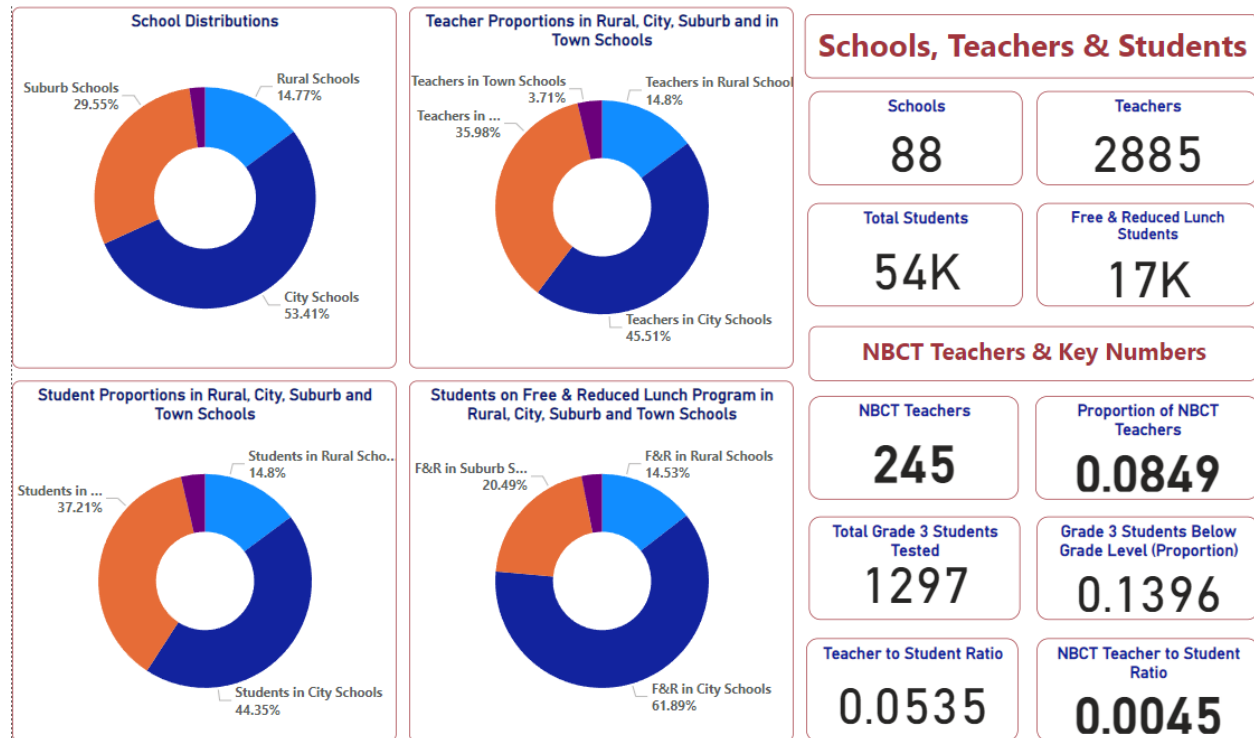


Figure 5.0

The chart also provides insights into the distribution of students across different types of schools in Madison. The highest proportion of students, as shown in the chart, is in city schools, followed by suburb schools. The percentage of students in rural and town schools is relatively low compared to the other areas. Those attending city schools are the majority at 44.35% while those that attend town schools are the lowest at 3.64%.

The chart provides insights into the distribution of students who come from economically disadvantaged backgrounds across different types of schools in Madison. The highest percentage of students on free and reduced lunch, as shown in the chart, is in city schools, followed by suburb schools. The percentage of students on free and reduced lunch in rural and town schools is

relatively low compared to the other areas. As shown, 61.89% of students in city schools are on free and reduced lunch, followed by 20.49% in suburb schools, 14.53% in rural schools, and 3.1% in town schools.

The highest proportion of students in Madison attend city schools (44.35%), followed by suburb schools. The percentage of students attending rural and town schools is relatively low compared to other areas, with town schools having the lowest proportion of students (3.64%). City schools in Madison have the highest percentage of students on free and reduced lunch (61.89%), followed by suburb schools (20.49%), rural schools (14.53%), and town schools (3.1%). The percentage of students on free and reduced lunch in rural and town schools is relatively low compared to other areas.

Relationship among Student Performance, Number of NBCT's & Socio-Economic Factors in Alabama

Student and Teacher Distributions

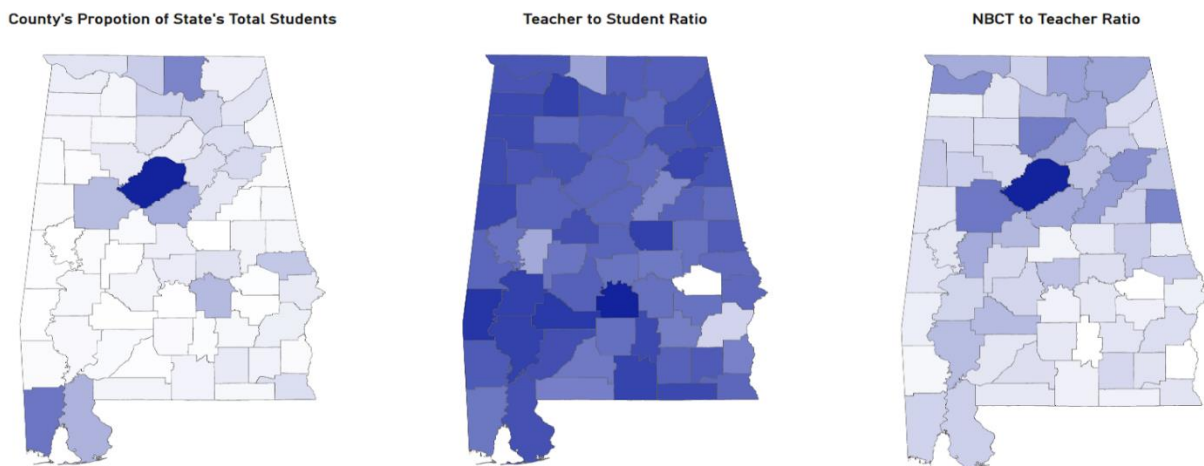


Figure 6.0

A summary on Student and Teachers Distribution in the above figure shows that the county with the highest proportion of the state's total students is Jefferson with 12.96%. The second highest is

Mobile with 8.09% followed by Madison with 7.22%. The county with the lowest proportion of students is Coosa with 0.11%. Some other counties with a low proportion of students include Greene, Choctaw, Lowndes, and Perry. It's worth noting that the total proportion of students across all counties adds up to 100%, which is expected since this is the entire population of students in the state of Alabama.

Moreover, the teacher to student ratio is calculated by dividing the total number of teachers in a county by the total number of students enrolled in that county. The state average for the teacher to student ratio is 3.47%, but this varies greatly by county. Madison County has a teacher-to-student ratio of 5.35%, which is slightly above the state average. This means that there are about 19 students for every teacher in the county. In addition, regarding the NBCTs to Teacher Ratio, the average NBCT's to Teacher Ratio for all counties is: 4.39% (calculated by dividing the number of NBCT by the number of teachers in each county).

The maximum NBCT's to Teacher Ratio is 19.9%, found in Jefferson County, and the minimum NBCT's to Teacher Ratio is 0, found in Crenshaw and Henry counties. As for Madison County, it has an NBCT's to Teacher Ratio of 8.5%, which is above the state average of 5.2%. This suggests that Madison County has a relatively higher proportion of teachers who are National Board Certified than many other counties in Alabama.

National Board-Certified Teachers (NBCT's) to Student Ratio

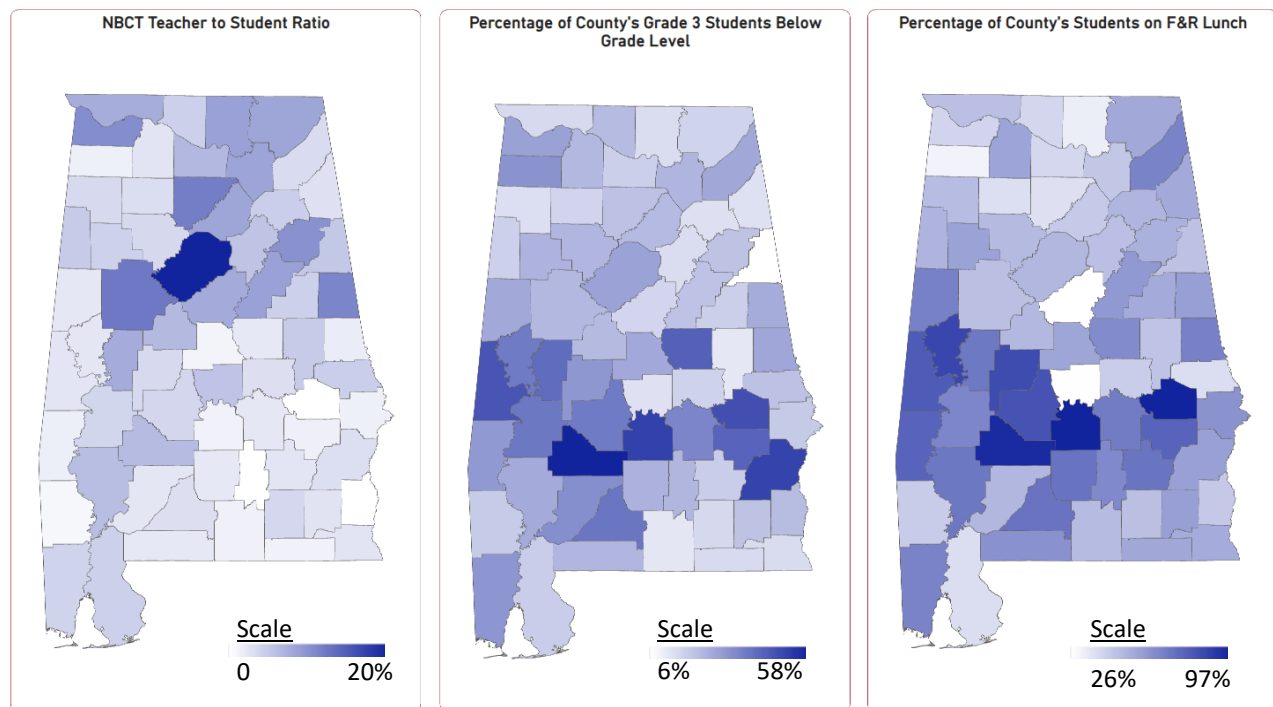


Figure 7.0

The first section from figure 7 shows the NBCT to Teacher Ratio for each county in Alabama. The NBCT to Teacher Ratio represents the proportion of NBCT's from all teachers in each county. The key findings in this section cover all the data we explored. The average NBCT to Teacher Ratio for all counties in Alabama is about 5.96%. The county with the highest NBCT to Teacher Ratio is Jefferson with 19.9%. The counties with the lowest NBCT to Teacher Ratios are Crenshaw and Henry Counties with 0 NBCTs. The county with the highest total number of NBCT's is Jefferson with 1,947 NBCT's, and the county with the lowest number of NBCTs is Crenshaw with 0 NBCT's. The total number of NBCT's in Alabama is 2,978, which corresponds to a total NBCT to Teacher Ratio of 2.98%. Based on these findings, it appears that there is variation in the distribution of NBCT's across the different counties in Alabama, which shows that the overall NBCT's distribution is not uniform.

Grade 3 Students Below Grade Level

The second section in this figure shows the percentage of Grade 3 students below grade level for each county in Alabama. The average percentage of Grade 3 students below grade level for all counties in Alabama is about 24.9%. The county with the highest percentage of Grade 3 students below grade level is Wilcox with 57.66%. The county with the lowest percentage of Grade 3 students below grade level is Cleburne with 5.88%. Shelby county has the lowest percentage of Grade 3 students below grade level among the top 10 most populous counties with approximately 15.7%. On the other hand, Madison County appears to have a relatively low percentage of grade 3 students below grade level and a high number of NBCT's compared to other counties in the state. This suggests that the county has a strong education system and is better equipped to help the students succeed academically.

The total number of Grade 3 students below grade level in Alabama is 39,161, which corresponds to a total percentage of Grade 3 students below grade level of 16.59%. Based on these findings, it appears that there is significant variation in the percentage of Grade 3 students below grade level across the different counties in Alabama.

Student Percentage Receiving Free and Reduced Lunch

From the third section of the figure, we found that, the percentage of county students receiving free or reduced lunch in Alabama varies from a high of 96.59% in Macon County to a low of 26.25% in Shelby County. The average percentage of students receiving free or reduced lunches across all counties in Alabama is 37.45%. The top 5 counties with the highest percentage of students receiving free or reduced lunch are Macon (96.59%), Lowndes (96.44%), Wilcox (94.18%), Greene (85.38%), and Perry (83.41%). On the other hand, the top 5 counties with the

lowest percentage of students receiving free or reduced lunch are Shelby (26.25%), Autauga (26.59%), Franklin (30.02%), Madison (31.76%), and Cullman (36.18%).

It is interesting to note that Madison County, which includes the city of Huntsville, has a percentage of students receiving free or reduced lunch that is lower than the state average, at 31.76%. Huntsville is home to several major employers, including NASA's Marshall Space Flight Center, the U.S. Army's Redstone Arsenal, and several high-tech companies, which may contribute to a higher average income for families in the area. Overall, the data shows that there is a wide range of economic diversity across the different counties in Alabama, with some counties having a large percentage of students receiving free or reduced lunch, while others have a much lower percentage.

Distribution of NBCT teachers in Alabama and other socio-economic factors

Based on the data analysis in the bubble chart below, the counties with the highest number of NBCT's are Jefferson, Madison, and Mobile. On the other hand, some counties, such as Bullock, Choctaw, and Wilcox, have a very low number of NBCT's. This indicates that the distribution of NBCT's across Alabama is uneven, with some areas having a greater concentration of NBCT's than others. Alongside this, the social-economic factor that we focused on was those students that are eligible for free and reduced lunch.

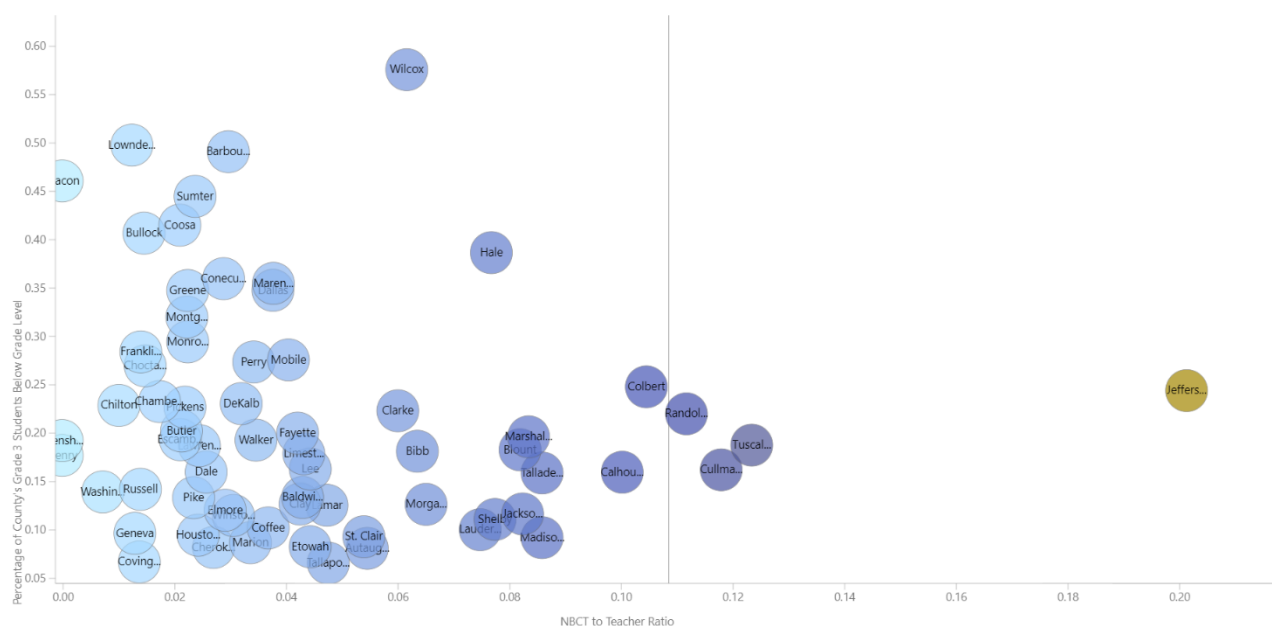


Figure 8.0

Further Insights on the Relationship among student performance, Number of NBCT's in Alabama and Other Socio-economic Factors

Relationship Between Poverty Levels and the Number of NBCT's in a given Area

There appears to be a negative correlation between poverty levels and the number of NBCT's in a given area. Counties with higher poverty levels tend to have a lower number of NBCT's, while counties with lower poverty levels tend to have a higher number of NBCT teachers. This suggests that poverty may be a factor that deters NBCT teachers from working in certain areas of Alabama.

Relationship between the Percentage of Students Receiving Free or Reduced-Price lunches and the Number of NBCT's in a given Area

There appears to be a negative correlation between the percentage of students receiving free or reduced-price lunches and the number of NBCT's in a given area. Counties with a higher percentage of students receiving free or reduced-price lunches tend to have a lower number of NBCT's, while counties with a lower percentage of students receiving free or reduced-price

lunches tend to have a higher number of NBCT's. This suggests that the socio-economic status of students may be a factor that affects the distribution of NBCT's in Alabama.

Relationship between Areas with Higher Levels of Educational attainment and the Percentage of NBCT's

There appears to be a positive correlation between areas with higher levels of educational attainment and the percentage of NBCT's. Counties with a higher percentage of residents with at least a bachelor's degree tend to have a higher percentage of NBCT's. This indicates that areas with a highly educated population may be more attractive to NBCT's.

Comparison between the Distribution of NBCT's and the Overall Distribution of Certified Teachers in Alabama

The distribution of NBCT's differs from the overall distribution of certified teachers in Alabama. While some counties have a similar percentage of NBCT's and certified teachers, others have a much higher or lower percentage of NBCT's compared to certified teachers. This suggests that the distribution of NBCT's is not proportional to the overall distribution of certified teachers in Alabama.

Relationship between Median Incomes and the Percentage of NBCT's

There appears to be a positive correlation between median incomes and the percentage of NBCT's. Counties with higher median incomes tend to have a higher percentage of NBCT's. This suggests that higher salaries may be a factor that attracts NBCT's to certain areas of Alabama.

Relationship between the Percentage of Students Performing at or above Grade Level in third Grade and the Percentage of NBCT's Teachers in a Given area

There appears to be a positive correlation between the percentage of students performing at or above grade level in third grade and the percentage of NBCT's in a given area. Counties with a higher percentage of students performing at or above grade level in third grade tend to have a

higher percentage of NBCT's. This suggests that the presence of NBCT's may have a positive impact on the academic performance of students.

Conclusion

Our results indicate that an investment by Huntsville/Madison County in increasing the proportion of NBCT's should therefore be strongly advocated for and supported.

Recommended Actions

It is recommended that C100 should focus on increasing the number of NBCT's in Madison County. C100 should also promote the incentivization of NBCT's certification among teachers to improve student performance. Based on the social economic factors therein, sponsoring the NBCT's in Madison County will ultimately improve student performance.

Recommendations for Further Research

- To assess the retention of NBCT's in schools, data on the number of NBCT's in schools over a period of about 3-5 years would be helpful.
- To assess the impact of retention of NBCT's in schools, data on county level performance profiles for multiple grades over a period of 3-5 years would be helpful.
- Below is one of the datasets that could be used for further analysis:
 - Alabama counties proficiency data on Grade 4/8/HS Reading/Math, graduation rates and post-school outcomes: <https://www.alabamaachieves.org/reports-data/student-data/special-education-reports/>

REFERENCES

Alabama Education Association. Retrieved from <https://www.myaea.org/>

Alabama State Data Center. Retrieved from <https://www.cba.ua.edu/asdc/>

Alabama State Department of Education. Retrieved from <https://www.alsde.edu/>

National Board for Professional Teaching Standards. Retrieved from <https://www.nbpts.org/>

National Center for Education Statistics. Retrieved from <https://nces.ed.gov/>

U.S. Census Bureau. Retrieved from <https://www.census.gov/>